

Lab03_PedroLacerda

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1)

```
library(readr)
```

Warning: pacote 'readr' foi compilado no R versão 4.5.3

```
library(dplyr)
```

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
```

Warning: pacote 'tidyr' foi compilado no R versão 4.5.3

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

```
-- Column specification -----
Delimiter: ","
chr (2): IATA_CODE, AIRLINE
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv("airports.csv",
                     col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

```
Rows: 322 Columns: 5
```

```
-- Column specification -----
Delimiter: ","
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
processa_bloco <- function(df, pos) {
  df %>%
    filter(!startsWith(as.character(DESTINATION_AIRPORT), "1")) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      qtd_voos = n(),
      .groups = "drop"
    )
}

flights_estatisticas <- read_csv_chunked(
  file = "flights.csv",
  callback = DataFrameCallback$new(processa_bloco),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- flights_estatisticas %>%
```

```
group_by(DESTINATION_AIRPORT) %>%
summarise(
  soma_total = sum(soma_atrasos),
  qtd_total = sum(qtd_voos),
  media_atraso = soma_total / qtd_total,
  .groups = "drop"
)

head(flights)
```

```
# A tibble: 6 x 4
  DESTINATION_AIRPORT soma_total qtd_total media_atraso
  <chr>                <dbl>      <int>      <dbl>
1 ABE                  12874        2220         5.80
2 ABI                   9077        2224         4.08
3 ABQ                 106415       18953         5.61
4 ABR                  -2227         657        -3.39
5 ABY                   7501         864         8.68
6 ACK                   2393         487         4.91
```

2)

```
aeroportos_ausentes <- anti_join(flights, airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

aeroportos_ausentes
```

```
# A tibble: 0 x 4
# i 4 variables: DESTINATION_AIRPORT <chr>, soma_total <dbl>, qtd_total <int>,
#   media_atraso <dbl>
```

```
flights <- left_join(
  x = flights,
  y = select(airports, IATA_CODE, CITY, STATE),
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)

head(flights)
```

```
# A tibble: 6 x 6
  DESTINATION_AIRPORT soma_total qtd_total media_atraso CITY      STATE
```

	<chr>	<dbl>	<int>	<dbl>	<chr>	<chr>
1	ABE	12874	2220	5.80	Allentown	PA
2	ABI	9077	2224	4.08	Abilene	TX
3	ABQ	106415	18953	5.61	Albuquerque	NM
4	ABR	-2227	657	-3.39	Aberdeen	SD
5	ABY	7501	864	8.68	Albany	GA
6	ACK	2393	487	4.91	Nantucket	MA

3)

```
aeroportos_por_estado <- airports %>%
  count(STATE, name = "quantidade_aeroportos", sort = TRUE)

aeroportos_por_estado
```

```
# A tibble: 54 x 2
  STATE quantidade_aeroportos
  <chr>           <int>
1 TX              24
2 CA              22
3 AK              19
4 FL              17
5 MI              15
6 NY              14
7 CO              10
8 MN               8
9 MT               8
10 NC              8
# i 44 more rows
```

4)

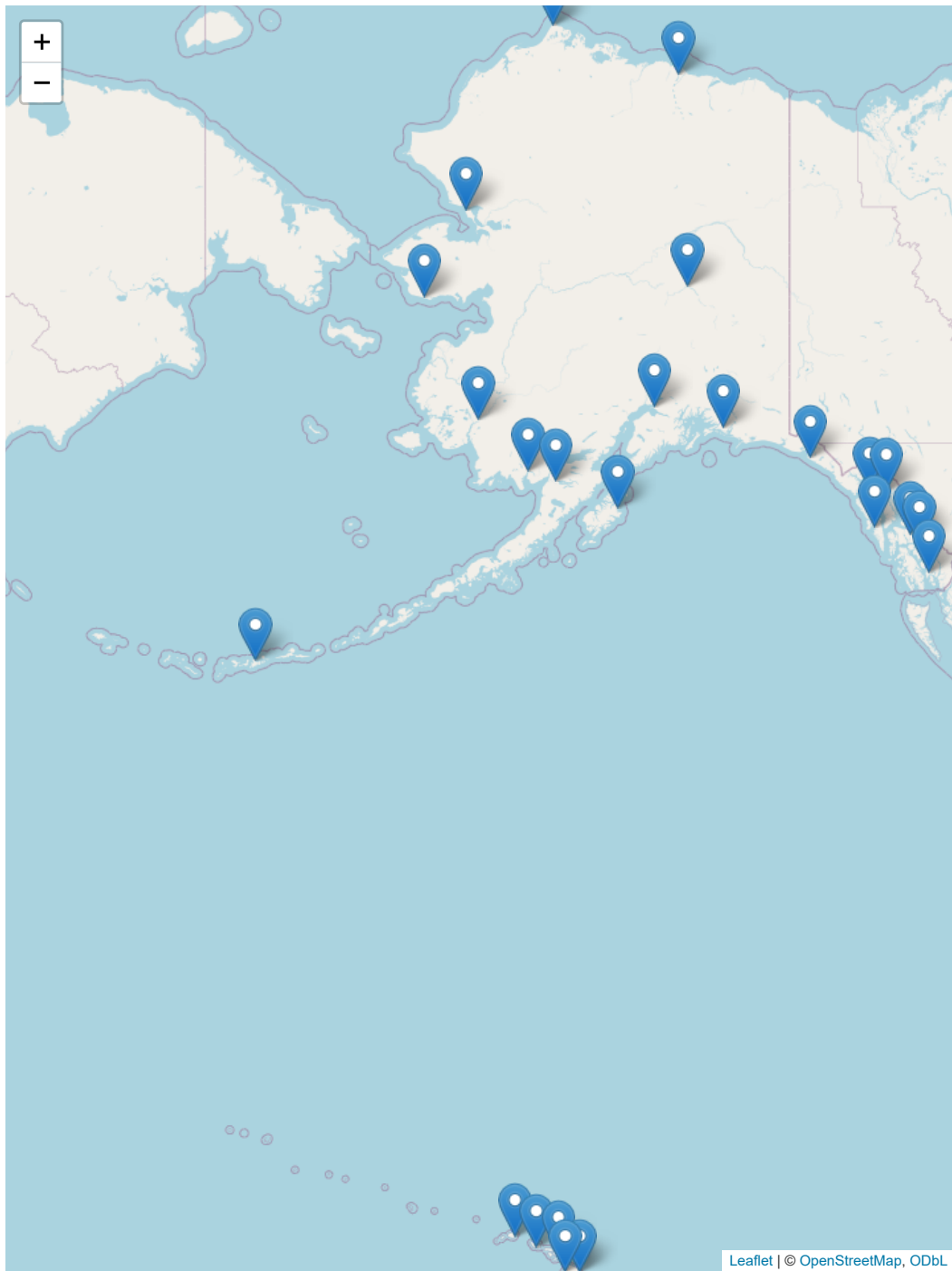
```
library(leaflet)
```

Warning: pacote 'leaflet' foi compilado no R versão 4.5.3

```
flights <- left_join(flights,
  select(airports, IATA_CODE, LATITUDE, LONGITUDE),
  by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

flights_mapa <- flights %>% drop_na(LATITUDE, LONGITUDE)
```

```
this <- leaflet(data = flights_mapa) %>%  
  addTiles() %>%  
  addMarkers(  
    lng = ~LONGITUDE,  
    lat = ~LATITUDE  
  )  
  
this
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 17:18:02 -03"
```